NOISEMAP · DESIGN PRECIS

NoiseMap design precis

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/02-concept.md @ ffa26f8		

REVISION HISTORY

Version	Date	Author	Change
0.7	2026-11-02	Amish Chadha	Kit 1.9.1 upgrade: the model is rebuilt at detail level D3 (89 parts, 3.45 kg); concept options and STEM check linked; design unchanged
0.6	2026-10-02	Amish Chadha	M12 pinout and first co-design partner to approach, decided on 2026-10-02 (NSM-DEC-001)
0.5	2026-10-01	Amish Chadha	Design made constructable (NSM-DDR-003); mass, cost and arm figures from NSM-CAL-001 v0.3; budget as a value-engineering target
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3; design choices adopted per NSM-DDR-001; numbers replaced by NSM-CAL-001 results; street pole adapter (BOM line 14) and parametric model added; FieldNode core at its TRL 3 figures
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, first-order numbers, safety, media)

0.1	2026-09-25	Amish Chadha	Initial scaffold

NoiseMap design precis

NoiseMap is a clamp-on street pole noise node. A digital MEMS microphone 4.0 m above the sidewalk feeds a small processor in the microphone head, which turns the sound into A- and C-weighted levels once a second and overwrites the audio within milliseconds. Only levels cross the cable to a standard FieldNode core, which sends a 22-byte record every 15 minutes over LoRaWAN. The TRL 3 calculations (NSM-CAL-001) give a design load of 12.2 mW, 53 days without sun and a range of 27.9 dBA to 119 dB SPL with the IM72D128 microphone fitted by default (34.9 dBA to 113 dB SPL with the ICS-43434). The NoiseMap parts are estimated at $72.00 against a $100 value-engineering target, which covers them alone; a full node with the $139.00 FieldNode core is $211.00. Made constructable under NSM-DDR-003, the node weighs 3.45 kg, within R10's 3.5 kg (relaxed from 3 kg under NSM-DDR-002). How to build it is in the prototype build plan, NSM-BLD-001. This design is option A of the concept options (NSM-OPT-001); the STEM check is NSM-STM-001.

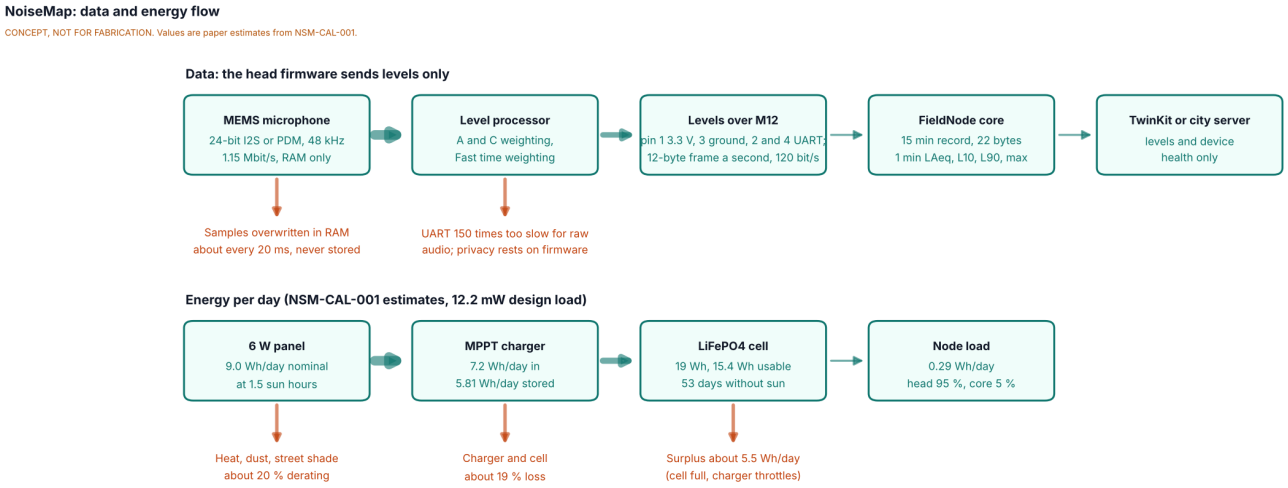


Seen from the front right and above, 24 deg elevation. Grey: Street pole, 114 mm, Sidewalk and road, Person, 1.75 m for scale

How it works

1. **Sense.** The MEMS microphone, port facing up under the head's acoustic port and inside a foam windscreen, streams 24-bit samples at 48 kHz to the level processor, over I²S from an ICS-43434 or PDM from an IM72D128.

2. **Weight and integrate.** The processor applies A and C weighting, a low-frequency equalizer for the microphone roll-off and a Z-weighted wind band below 40 Hz as digital filters, then computes each second LAeq,1s, the Fast time-weighted maximum and minimum (LAFmax, LAFmin), LCeq,1s and the wind band level. Samples live in a small RAM buffer that is overwritten about every 20 ms. The load is about 13 % of a 48 MHz core.
3. **Hand over levels only.** Once a second the head sends a 12-byte frame to the FieldNode core over a 9,600 baud UART in the M12 cable. The head has no data storage and no radio.
4. **Summarize.** The FieldNode core builds a 15-minute record: fifteen 1-minute LAeq values, the 15-minute LAeq, LAFmax and LAFmin, L10 and L90 (from the 900 one-second values), LCeq and a status byte with the wind flag, 22 bytes in all.
5. **Send.** The core sends each record over LoRaWAN to the lab's TwinKit gateway, a city server or The Things Network, and stores it in flash with a time stamp if the link is down. At SF10 and slower the core lengthens the interval automatically to stay within The Things Network's 30 s a day: 24 min at SF10, 48 min at SF11, 87 min at SF12; the record stays 22 bytes, and its fifteen sub-interval LAeq values each span a fifteenth of the interval (NSM-DDR-002). Night indicators (Lnight, Lden) and the rain flag are computed on the server.



Main components

Table 1. Main components. Numbers match the exploded view (Figure 3), `cad/src/model.py` and `bom/bom.csv`. Lines 1 to 6 are the FieldNode core, costed in the FieldNode repo.

#	Component	Choice	Notes
1	FieldNode enclosure	IP65 polycarbonate box 150 x 90 x 200 mm with vent and glands	From FieldNode, unchanged
2	FieldNode power and radio board	MPPT charger with cold-charge lockout, STM32WL-class LoRaWAN module, SPI flash, whip antenna, two M12 ports	From FieldNode, unchanged
3	LiFePO4 cell	One 3.2 V, 6 Ah cell	FieldNode standard; 53 days at the NoiseMap load
4	Solar panel	6 W, 290 x 200 mm, 40° tilt, doubling as a hood	FieldNode standard
5	Panel tilt bracket	Flat-bar posts and struts	From FieldNode
6	FieldNode pole mounting kit	Back plate 180 x 320 x 3 mm	Only the plate is used on street poles
7	Microphone arm	25 x 2 mm aluminum tube, 383 mm, in a bought tube flange on a saddle bent from 2.5 mm aluminum sheet, held by its own two band clamps	Puts the microphone 0.45 m from the pole face; turns independently of the core
8	Microphone head housing	Printed ASA tube 40 mm OD x 150 mm, 3 mm acoustic port, drip skirt, hydrophobic membrane	NSM-DDR-001 D13
9	MEMS microphone	Adapter board for an ICS-43434 (I²S, 65 dB SNR, TDK InvenSense) or an IM72D128 (PDM, 72 dB(A) SNR, DNMS)	D3; the ICS-43434 is listed end of life
10	Level processor	Cortex-M4F class low-power board (STM32L4 class) with I²S and PDM inputs and read-out protection	D4
11	Windscreen and bird spike	90 mm open-cell foam ball bored 41 mm, stainless spike	Replace about yearly (estimate)
12	Sensor cable	M12 5-pin, about 1.5 m, 3.3 V, ground and UART	FieldNode port: pin 1 at 3.3 V, pin 3 ground, pins 2 and 4 the UART pair, on FieldNode's candidate pinout, which NoiseMap asks FieldNode to adopt (decided 2026-10-02)
14	Street pole adapter	Two 90° aluminum V-blocks 112 x 63 x 16 mm sawn from bar, on the FieldNode back plate's own screw holes, and two long stainless bands through its slots	Seats 60 to 140 mm poles; replaces FieldNode's 40 to 60 mm V-blocks

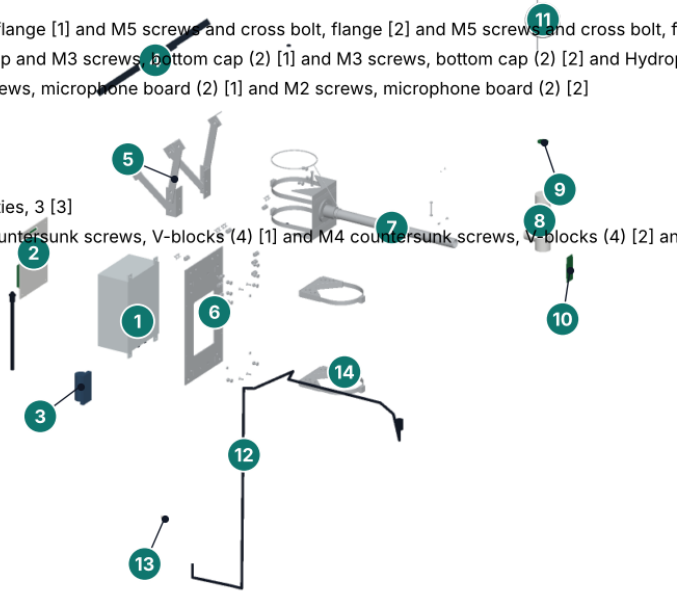
Line 13 of the BOM (fasteners, safety lanyard, ties, tape) has no callout.

NoiseMap: exploded view

github.com/BoujeeEnjinia1701/noisemap

CONCEPT, NOT FOR FABRICATION

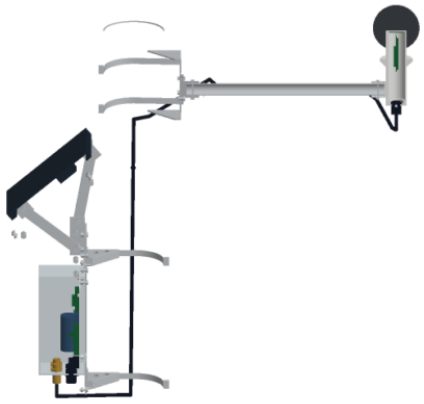
1 FieldNode enclosure body and FieldNode enclosure lid and FieldNode enclosure lugs (4) and FieldNode m5 screws and nuts, lugs (4) [1] and FieldNode n
2 FieldNode sensor ports, 2 x M12 and FieldNode antenna, bulkhead and whip and FieldNode internal mounting plate and FieldNode m4 screws, internal pl
3 FieldNode liFePO4 cell in fused holder
4 FieldNode solar panel, 6 W and FieldNode m4 bolts, panel clips (8) [1] and FieldNode m4 bolts, panel clips (8) [2] and FieldNode m4 bolts, panel clips (8
5 FieldNode plate clip, right and FieldNode plate clip, left and FieldNode post, right and FieldNode post, left and FieldNode strut, right and FieldNode strut,
6 FieldNode back plate
7 Arm saddle and Tube flange (bought) and M5 screws and cross bolt, flange [1] and M5 screws and cross bolt, flange [2] and M5 screws and cross bolt, f
8 Head housing and M4 cross bolt, head socket [1] and Head bottom cap and M3 screws, bottom cap (2) [1] and M3 screws, bottom cap (2) [2] and Hydro
9 MEMS microphone on adapter board and Acoustic gasket and M2 screws, microphone board (2) [1] and M2 screws, microphone board (2) [2]
10 Level processor board
11 Foam windscreen and Bird spike
12 Cable gland, M16 and Sensor cable
13 Port label, 3.3 V and Cable ties, 3 [1] and Cable ties, 3 [2] and Cable ties, 3 [3]
14 Street pole V-block, lower and Street pole V-block, upper and M4 countersunk screws, V-blocks (4) [1] and M4 countersunk screws, V-blocks (4) [2] an



Seen from the front right and above, 24 deg elevation; numbers match bom/bom.csv

NoiseMap: cutaway github.com/BoujeeEnjinia1701/noisemap

CONCEPT, NOT FOR FABRICATION



Front half removed; seen from the front and above, 18 deg elevation

The general arrangement is drawing NSM-DWG-001 Rev P3 (`cad/drawings/NSM-DWG-001.pdf`), generated from `cad/src/model.py` , which also exports STEP and STL files in `cad/step/` and `cad/stl/` .

Key numbers

All values come from NSM-CAL-001 and are paper estimates.

Table 2. Key numbers.

Quantity	Value	Requirement
Microphone self-noise	29.0 dBA (ICS-43434); 22.0 dBA (IM72D128)	
Measuring range	34.9 dBA to 113 dB SPL (ICS-43434, 103 to 108 dBA for traffic); 27.9 dBA to 119 dB SPL (IM72D128, 109 to 114 dBA)	R4 at risk; IM72D128 fitted by default
Response at 63 Hz	-2.8 dB before equalizing; ±0.8 dB after, with ±20 % corner spread	R5 at risk
Acoustic port resonance	36.2 kHz; +0.44 dB at 8 kHz	R5
Expanded uncertainty	2.9 dB on assumed terms; facade bias up to 2.4 dB	R3 not met on paper
Processor	132 cycles per sample, 13.2 % of 48 MHz, 2.15 mA	
Node load from the cell	12.2 mW, 0.29 Wh/day (22.6 mW upper bound)	12 % of FieldNode's 100 mW allowance
Harvest at 1.5 peak sun hours	5.81 Wh/day stored	R7 met on paper
Autonomy	53 days; 37 days at -20 °C	R7 met on paper
Record and airtime	22 bytes; 23.7 s/day at SF9 at 15 min; at most 30.0 s/day at SF10 to SF12 with the interval rule	R2 met; R12 met on paper
Link capacity	UART 7,680 bit/s (150 times too slow for raw audio, enough for a speech codec)	R1 rests on firmware
Mass on the pole	3.45 kg for the constructable design	R10 met on paper at 3.5 kg; margin 0.05 kg
Wind at 35 m/s	Arm, head and windscreen 16 N; arm stress factor 24; clamp twist factor 4.2	R10 met on paper
Parts cost	NoiseMap $72.00 against a $100 value-engineering target; FieldNode core $139.00; full node $211.00	R13 met on paper

The accuracy of the measurement depends on the membrane, the windscreen, diffraction round the head and reflections from nearby facades, none of which can be settled on paper.

Key design choices

All of these are decided by Amish, 2026-09-25: go with recommendation (NSM-DDR-001, NSM-DDR-002).

- **Levels, never audio (D9).** The only data that leave the microphone head are levels once a second. The head has no data storage and no radio, and the firmware has no code path that writes or sends samples.
- **Where privacy comes from (D2).** The 9,600 baud cable is 150 times too slow for raw audio, but fast enough for a speech codec such as Codec 2 (700 to 3,200 bit/s), and the radio could carry about 31 s of such speech a day under fair use. The guarantee therefore rests on open, auditable head firmware with read-out protection and a published build hash. Public copy never claims privacy "by physics".

- **Processing in the head (D4).** Keeping the audio next to the microphone means the M12 cable only ever carries levels.
- **Standard FieldNode core (D10).** At 12.2 mW the node uses 12 % of FieldNode's allowance with its standard 6 W panel and one cell, and stays in credit on the hot clear days that stop FieldNode charging.
- **Microphone 4.0 m above the sidewalk, 0.45 m from the pole, pointing up (D11).** This matches noise-mapping practice; the pole adds only 0.25 dB by reflection. The arm has its own clamp, so the core can face the equator while the arm faces the street.
- **A and C weighting (D12)** with 15-minute records and 1-minute LAeq (D5).
- **Microphone part (D3).** A small adapter board takes either the ICS-43434 or the IM72D128; the IM72D128 gives 7 dB more margin at the quiet end.
- **Field calibration (D7).** Each node is checked with a shared 94 dB, 1 kHz calibrator at install and at each windscreen change, and compared with a class 1 reference meter at one site per deployment, which also sets the site's reflection correction.
- **Public notice (D8)** on each pole saying what is measured, with a link to this repository.
- **R10 at 3.5 kg (O2).** Under NSM-DDR-003 the V-blocks are sawn from 16 mm bar with drilled holes and the arm saddle is bent from sheet; the node is 3.45 kg, so the 0.05 kg margin is thin and any added part (an anemometer, a heavier windscreen) needs a mass check.
- **Level-based wind flag (O3).** Wind is flagged from the Z-weighted band below 40 Hz and its one-second spread, with no added sensor; rain is flagged on the server from weather data. The threshold needs a field comparison with an anemometer, which is TRL 4 work and on hold.
- **Interval rule at slow data rates (O4).** The core lengthens the reporting interval automatically at SF10 to SF12, following FieldNode's interval item, so R12 is met at every spreading factor.

Safety

> **Safety:** Installing on a street pole is work at height next to traffic. Install only with the pole owner's permission, by trained crews, with fall protection and traffic management as local rules require. Keep clear of overhead power lines and street-light wiring, and never open a pole's electrical hatch. The node adds about 96 N of wind load at 3.5 to 4 m in a 35 m/s gust; the pole owner should confirm the pole can take it.
>
> **Safety:** The node contains a LiFePO4 cell of about 19 Wh. Fuse the cell, charge only within the cell maker's temperature limits (FieldNode's cold-charge lockout applies), and do not install a node with a swollen or damaged cell.
>
> **Safety:** A falling part from 4 m can injure people below. Fit a secondary safety lanyard to the microphone arm, tighten the bands to a stated torque (the clamp margins in NSM-CAL-001 assume 1,000 N preload), and check the clamps after the first storm.
>
> **Safety:** Do not test the node near its upper range with loudspeakers or calibrators at close range without hearing protection.

Privacy. No images, audio recordings or personal identifiers leave the device; only sound levels are stored or sent. The guarantee rests on the published head firmware and its build hash. Check local data protection law before any deployment, and put a notice on the pole saying what is measured and where the design is documented.

Use of the data. NoiseMap readings are indicative. They are not certified measurements and must not be presented as legal evidence without a co-located certified instrument.

Open questions

- First co-design partner and street (NSM-DDR-001 O1): decided 2026-10-02; the first candidate to approach is a city environmental noise team or a university acoustics group that can put a Class 1 reference sound level meter beside the node on the test street; a Sensor.Community group is a useful second contact for volunteer hosts. Not yet agreed.
- Beyond TRL 3, on hold: measure the response of the chosen microphone in the printed head with membrane and windscreen, the windscreen insertion loss, the wind-band threshold against an anemometer, and the site reflection correction.

Concept media: blueprint sheet, interactive 3D model.